Israel Souza Ferreira

Applied AI / Research Engineer

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Professional Summary

Computer Science graduate contributing to multimodal ML R&D: representation learning, vector retrieval, multi-task deep learning, and LLM components with DSPy. Experience implementing research ideas and comparing approaches through metrics and baselines.

Professional Experience

Tieta.ai · Machine Learning R&D · Remote

R&D Researcher · Jul 2026–present

ML Researcher · CNPq RHAE Fellow (part-time) · Apr 2025–Jun 2026

- Collaborated on implementing and experimenting with multi-task deep learning architectures, combining multimodal representations for regression and classification tasks.
- Contributed to developing and evaluating multimodal representations and retrieval using embeddings, vector search, and similarity measures, comparing text and image representations in vector spaces.
- Contributed to experiments with LLM and DSPy components for structured information extraction and semantic representation construction.
- Collaborated on hypothesis testing, baseline implementation, and approach comparisons using regression and classification metrics, drawing on scientific literature to guide experiments and modeling decisions.

Independent consulting · Educational institution · Mar 2026–present

Full Stack Developer and ML Engineer · Face Clock Evoluir

v1.3.0-rc.3: supervised institutional pilot candidate deployed to a test environment.

- Built 1:N facial identification, FastAPI APIs, and PostgreSQL/pgvector persistence for a CPU-only server; fixed concurrent extraction failures.
- Recalibrated the facial threshold: unknown-person acceptance fell from 45.4% to 0.2% in a controlled test. Calibration on the institution’s population remains pending.
- Used benchmarks to retain exact search and reduce frames; migrated document-classifier inference to ONNX Runtime with decision-parity checks on the held-out test.

Research and Projects

Triple Roman Domination in Graphs · Undergraduate thesis · 2024–2025

- Identified a counterexample to a published ILP formulation; implemented a corrected model, GA, and ACO (C++, Python/Pyomo). Evaluated 362 graphs; one run per instance limits conclusions.

Technical Skills

Python, PyTorch, scikit-learn; multimodal and multi-task deep learning, representation learning, embeddings, vector search, LLMs, DSPy.

FastAPI, PostgreSQL/pgvector, Docker, ONNX Runtime, Git, pytest; SQL, Pandas, NumPy.

Cross-validation, calibration, ablation, classification metrics; C++, GA, ACO, Python/Pyomo.

Education

B.Sc. in Computer Science · Federal University of Ceará (UFC), Quixadá, Brazil

2021–19 Aug 2026 · Thesis defended in Feb 2025.

Languages

Native Portuguese. English: good communication; fluent technical reading of papers and documentation.